

MX-5 NB turbo run sheet

Calibration: **18Jul_boost_pwm_limits_v7.mecal** Date: _____ Ambient: _____ °C Fuel: 98 RON

Do Part A in the driveway first. Part B needs the car moving; keep the boost pulls (B4, B5) for a closed road, track day or the dyno. Start a fresh log for each test and note the log number in the box.

Before you start

- ☐ Save a copy of the calibration currently in the car (rollback).
- ☐ Open v7 in MEITE and spot-check: dead time axis shows 12.5 / 13.25 / 13.5 / 13.75 V cells; lean protection triggers on *leaner* than target; oil protection on with the 130–240 kPa table.
- ☐ Enable logging of: knock cyl 1–4 peak and event count, knock reading peak, knock add, overrun status, extra idle load switch, fan status, idle target / error / P / I, boost P / I / CL init duty, EPS status channels.
- ☐ Burn v7. Check the actuator and solenoid hoses and clamps.

Part A: driveway (stationary)

A1. Cold start: IAT check

- ☐ Before starting a cold engine, compare IAT reading to ambient. IAT: _____ °C, ambient: _____ °C (should be within 1–2 °C).
- ☐ Note the IAT sensor body: open element / brass closed: _____

A2. Warm up, then knock sensor tap test

- ☐ Warm fully (coolant 85 °C+). Note when the fan cycles (on 92 °C, off 89 °C).
- ☐ At idle, tap the block near the knock sensor lightly with a spanner. Knock reading jumps? Y / N

A3. Power steering switch

- ☐ At warm idle, turn the wheel to full lock and hold. Extra load switch status changes? Y / N. Idle duty rises ~6%? Y / N
- ☐ If it never changes: check switch and wiring. If it is stuck on: active state is inverted.

A4. Idle voltage test (dead times)

- ☐ Temporarily disable closed loop lambda. Fan off. Wheel straight.
- ☐ Log ~2 min while switching headlights, rear demister, then all together, 20 s each. Watch AFR vs battery voltage.
- ☐ AFR steady as voltage moves = good. Leaner when voltage drops = low-voltage dead times too short; richer = too long.
- ☐ Re-enable closed loop lambda.

Log no: _____

A5. Idle valve duty sweep

- ☐ Fan off, wheel straight, closed loop lambda on. Set idle to manual duty (or edit the warm cell live). Hold each step ~15 s, then record. Stop and raise duty if rpm falls below ~850.

Duty %	30	27	25	23	21	19	17
RPM							
MAP kPa							
AFR							

If rpm stops falling below some duty: extra air is setting the floor (throttle stop, idle air screw, vacuum leak, BOV). Restore open loop idle afterwards. Log no: _____

A6. Idle spark scatter (optional, do last)

- ☐ Set spark scatter to about +3° at –300 rpm error and –3° at +300 rpm (0 at 0). Confirm the sign: with rpm below target the ECU must *add* advance.
- ☐ Blip the throttle a few times; return to idle should be smoother, not oscillating. If it hunts, reduce or remove.

Part B: on the road

B1. Overrun and return to idle

- ☐ Six to eight decels from 3,000–4,000 rpm to idle in gear, throttle fully closed; then three with the clutch in.
- ☐ Feel for droop or stumble on the way to idle. Lowest rpm seen: _____. Any stall? Y / N

Expected: fuel cut on decel (overrun status), fuel back in at 1,800 rpm, lambda trim not winding up to +20%. Log no: _____

B2. Knock baseline (off boost)

- ☐ Steady moderate-throttle cruising in 3rd and 4th across as much of 1,500–6,000 rpm as is legal, staying below ~95 kPa.
- ☐ Knock add should stay at 0. Any knock retard during gentle cruise = threshold too low, note rpm: _____

Log no: _____

B3. Knock reproduction

- ☐ Pull away gently in 1st, three times.
- ☐ Moderate-throttle roll-on in 3rd from ~1,500 rpm into light boost, three times.
- ☐ Note any knock retard and which cylinder peaks rise: _____

Log no: _____

B4. Wastegate solenoid unplugged (closed road / track)

- ☐ Unplug the solenoid. One full-throttle pull in 3rd from ~2,500 rpm.
- ☐ Peak MAP: _____ kPa. Should be spring pressure only (around 150 kPa). If it keeps rising: lift, and check solenoid plumbing.

Over-boost cut (225 kPa) is the backstop. Log no: _____

B5. Open loop boost duty steps (closed road / track / dyno)

- ☐ Boost mode = Open Loop, flat duty table. For each step set over-boost cut ~20 kPa above the previous step's peak. Full-throttle 3rd-gear pull ~2,500–6,500 rpm. Stop increasing duty once peak reaches ~190 kPa.

Duty %	Cut set (kPa)	Peak MAP (kPa)	RPM at peak	Knock add? (Y/N)	Log no
0					
20					
30					
40					
50					

Afterwards: set boost mode back to Closed Loop and over-boost cut back to 225 kPa.

Abort immediately (lift, note the log time) if

- Oil pressure warning or the 2,000 rpm oil protection limit trips.
- Lean protection trips (3,000 rpm limit), or AFR goes above ~12.5 on boost.
- MAP overshoots the expected value by more than ~15 kPa, or the over-boost cut fires unexpectedly.
- Knock retard reaches more than ~3° repeatedly in the same area.
- Coolant above ~100 °C or oil above ~120 °C.

After the drive

- ☐ Copy all logs off the SD card; note which log matches which test.
- ☐ Upload the logs, the calibration that was in the car, and filled-in copies of A5 / B4 / B5 to the next chat with the worklog link.